Gas Laws

61. A gas at $27\,℃$ temperature and 30 atmospheric pressure is allowed to expand to the atmospheric pressure. If the volume becomes 10 times its initial volume, then the final temperature becomes

(a) 100°*C* (b) 173°*C*

(c) 273°*C* (d) – 173°*C*

62. When volume of system is increased two times and temperature is decreased half of its initial temperature, then pressure becomes

(a) 2 times (b) 4 times

(c) $\frac{1}{4}$ times (d) $\frac{1}{2}$ times

63. A balloon contains 1500 m^3 of helium at $27°C$ and 4 atmospheric pressure. The value of helium at $-3°C$ temperature and 2 atmospheric pressure will be

(a) 1500 m^3 (b) 1700 m^3

(c) 1900 m^3 (d) 2700 m^3

64. In the relation $n = \frac{PV}{RT}, n =$

(a) Number of molecules

(b) Atomic number

(c) Mass number

(d) Number of moles

65. A gas at 27°*C* has a volume *V* and pressure *P*. On heating its pressure is doubled and volume becomes three times. The resulting temperature of the gas will be

(a) 1800°*C* (b) 162°*C*

(c) 1527°*C* (d) 600°*C*

66. The equation of state corresponding to 8 *g* of O_2 is

(a) *PV* = 8*RT* (b) *PV* = *RT*/4

(c) *PV* = *RT* (d) *PV* = *RT*/2

67. The equation of state for 5 *g* of oxygen at a pressure *P* and temperature *T*, when occupying a volume *V*, will be

(a) $PV = (5/32)RT$

(b) $PV = 5RT$

(c) $PV = (5/2)RT$

(d) $PV = (5/16)RT$

(Where *R* is the gas constant)

68. O_2 gas is filled in a vessel. If pressure is doubled, temperature

becomes four times. How many times its density will become

(a) 2 (b) 4

(c) $\frac{1}{4}$ (d) $\frac{1}{2}$

69. The volume of a gas will be double of what it is at 0°*C* (pressure remaining constant) at

(a) 1092 *K* (b) 273 *K*

(c) $546°C$ (d) $273°C$

70. The pressure and temperature of two different gases is *P* and *T* having the volume *V* for each. They are mixed keeping the same volume and temperature, the pressure of the mixture will be

(a) *P* / 2 (b) *P*

(c) 2*P* (d) 4*P*

71. Three containers of the same volume contain three different gases. The masses of the molecules are m_1, m_2 and m_3 and the number of molecules in their respective containers are N_1, N_2 and N_3. The gas pressure in the containers are P_1, P_2 and P_3 respectively. All the gases are now mixed and put in one of the containers. The pressure *P* of mixture will be

(a) $P < (P_1 + P_2 + P_3)$

(b) $P = \frac{P_1+P_2+P_3}{3}$

(c) $P = P_1 + P_2 + P_3$

(d) $P > (P_1 + P_2 + P_3)$

72. At a given volume and temperature, the pressure of a gas

(a) Varies inversely as its mass

(b) Varies inversely as the square of its mass

(c) Varies linearly as its mass

(d) Is independent of its mass

73. The rate of diffusion is

(a) Faster in solids than in liquids and gases

(b) Faster in liquids than in solids and gases

(c) Equal to solids, liquids and gases

(d) Faster in gases than in liquids and solids

74. A vessel is filled with an ideal gas at a pressure of 10 atmospheres and temperature 27°*C*. Half of the mass

of the gas is removed from the vessel and temperature of the remaining gas is increased to 87°*C*. Then the pressure of the gas in the vessel will be

(a) 5 *atm* (b) 6 *atm*

(c) 7 *atm* (d) 8 *atm*

75. When a gas in a closed vessel was heated so as to increase its temperature by 5^oC, there occurred an increase of 1% in its pressure. The original temperature of the gas was

(a) 500^oC (b) 273^oC

(c) 227^oC (d) 50^oC

76. Two thermally insulated vessels 1 and 2 are filled with air at temperatures (T_1, T_2), volume (V_1, V_2) and pressure (P_1, P_2) respectively. If the valve joining the two vessels is opened, the temperature inside the vessel at equilibrium will be

(a) $T_1 + T_2$

(b) $(T_1 + T_2)/2$

(c) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_2+P_2V_2T_1}$

(d) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_1+P_2V_2T_2}$

77. For matter to exist simultaneously in gas and liquid phases

(a) The temperature must be 0 *K*

(b) The temperature must be less than 0^oC

(c) The temperature must be less than the critical temperature

(d) The temperature must be less than the reduced temperature

78. The value of critical temperature in terms of Vander Waal's constant *a* and *b* is

(a) $T_c = \frac{8a}{27Rb}$ (b) $T_c = \frac{a}{2Rb}$

(c) $T_c = \frac{8}{27Rb}$ (d) $T_c = \frac{27a}{8Rb}$

79. In Vander Waal's equation *a* and *b* represent

$$\left(P + \frac{a}{V^2}\right)(v - b) = RT$$

(a) Both *a* and *b* represent correction in volume

(b) Both *a* and *b* represent adhesive force between molecules

(c) *a* represents adhesive force between molecules and *b* correction in volume

(d) *a* represents correction in volume and *b* represents

adhesive force between molecules

80. The gas in vessel is subjected to a pressure of 20 atmosphere at a temperature 27°*C*. The pressure of the gas in a vessel after one half of the gas is released from the vessel and the temperature of the remainder is raised by 50°*C* is

(a) 8.5 *atm* (b) 10.8 *atm*

(c) 11.7 *atm* (d) 17 *atm*

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